Fly-in-fly-out work: A review of the impact of an extreme form of work-related travel on mental health

Fruhen, L.S.; Gilbert, J.M.; Parker, S.K.

2023, Article / Letter to editor (Organizational Psychology Review, 13, 2, (2023), pp. 177-204)
Doi link to publisher: https://doi.org/10.1177/20413866221134938

Special Issue: Planes, Trains, and Automobiles: Commuting in the 2020s and Beyond

Organizational Psychology Review

Fly-in-fly-out work: A review of the impact of an extreme form of work-related travel on mental health

Organizational Psychology Review
2023, Vol. 13(2) 177–204

DOI: 10.1177/20413866221134938
journals.sagepub.com/home/opr
SAGE

Laura S. Fruhen
School of Psychological Science, The University of Western Australia, Perth, Australia

Jess Gilbert
Centre for Transformative Work Design, Future of Work Institute, Curtin University, Perth, Australia

Sharon K. Parker
Centre for Transformative Work Design, Future of Work Institute, Curtin University, Perth, Australia

Abstract
Large distances between work and home require many workers to stay away from home for work over extended periods. An extreme case of such work is fly-in-fly-out (FIFO) work. FIFO work requires workers to stay, over a fixed number of days or weeks, in remote employer-arranged accommodation. Given the disruptive nature of this work arrangement, it is important to develop an understanding of its implications for worker mental health. Based on a systematic review, we identify mostly negative mental health implications of FIFO work and propose an integrative model that brings together FIFO work's key features: the simultaneous fracturing and blending of personal and work lives. The model can guide future research. For example, we suggest that researchers investigate how FIFO workers and other work-related travelers experience fracturing and blending, and how they manage the frequent fluctuation between these two extremes.

Plain language summary
Fly-in-fly-out work represents a specific and extreme case of work-related travel in which workers are housed by employers and have fixed schedules that prescribe blocks of time at work

Paper received 1 March 2021. Received revised May 15, 2022. Accepted October 1, 2022

Corresponding author:
Laura Fruhen, School of Psychological Science, The University of Western Australia (M304), 35 Stirling Highway, 6009 Perth, Australia.
Email: laura.fruhen@uwa.edu.au

followed by blocks of time at home, alongside other rules. This paper provides an overview of the literature regarding the mental health implications of FIFO work. Moreover, an integrative model of FIFO work is proposed that brings together its key defining attributes: the simultaneous fracturing and blending of personal and work lives, which FIFO workers, and by extension other work-related travellers, experience. The model identifies the key attributes of this type of work arrangement that shape fracturing namely the commute (i.e., duration, mode, distance) and rosters (i.e., length and ratio of time on and off) and blending experiences, namely accommodation (i.e., type and quality), extent to which autonomy is limited (i.e., via choice in food, activities, scheduling) and social isolation. While developed for an extreme, highly standardized, and specific case of work-related travel, the model may also be useful in research focused on work related travel more broadly,

Keywords
roster, business travel, overnighting, working away from home, work design

An overall trend towards greater mobility means that workers cover greater distances and travel more frequently for work (Rüger et al., 2011). One consequence of this higher general mobility, as well as the value that increased mobility may have in terms of flexibility to fulfil workforce demands, is an increase in the already high number of people who spend significant amounts of time away from home for work (Costa et al., 2006; Gustafson, 2012; Montazer et al., 2020). Statistics for the European Union (EU) show that in 2019, EU residents made 125 million trips for business purposes, with the majority of these trips requiring them to stay away from home for up to three nights (73.8%; Eurostat, 2021). For the U. S., annually, 405 million business trips are taken and the Department of Transportation estimates that 7% of these trips involved workers travelling distances of 1,000 miles or more (Lake, 2020). Increases in the amount of time people spend being away from home for work have been predicted for the future including a recovery following Covid-19 restrictions (Caputo et al., 2021).

A specific form of work-related travel, which we focus on in this article, is fly-in-fly-out (FIFO) work. The practice of FIFO work, prevalent in contexts such as the physical resources sector (e.g., mining, oil and gas), involves employees being flown to remote locations to carry out their work (Government of Western Australia, Department of State Development, 2016). FIFO work has three key attributes (Storey, 2001; Vojnovic et al., 2014; Henry et al., 2013). First, FIFO work is characterized by the place of work being so far removed from the workers' usual residence that it is not possible for them to return home at the end of a shift. Consequently, FIFO work usually involves long distance travel, often in workers' own time (i.e., during their time off). Second, FIFO work typically involves carrying out a fixed work schedule over several days, which is followed by a fixed break period. This type of work arrangement means workers experience prolonged absences from friends and families, commonly between five and 28 days. Third, FIFO work involves workers usually being provided by employers with food and drinks, and accommodation close to the worksite.

In this review, we focus on FIFO work as a specific form of work-related travel. We seek to develop a theoretical model that can capture FIFO work and its impact on workers. Doing so is important because FIFO work has attracted a great deal of attention from the media, researchers, policymakers, unions, and employers, mostly because of concerns around the mental health of FIFO workers. Investigating mental health in the context of FIFO work is particularly important, given that FIFO workers, in terms of their age and gender can be considered an at-risk group for mental health issues and suicide (Australian Bureau of Statistics, 2016). Although the exact number of FIFO employees fluctuates with commodity prices and other economic factors (a phenomenon described by Peck (2013, p. 250), as "the FIFO faucet being adjusted on a continuous basis"), FIFO work is relatively common in countries such as Australia, Canada and Russia that have large distances and industries connected with mining and related occupations. For example, in Western Australia FIFO work provides employment for an estimated 60,000 people, constituting 4.78% of the workforce[1], with estimates that almost 10% of the population are directly impacted by this form of work (Education and Health Standing Committee on FIFO worker mental health, 2015).

In what follows, we discuss the research on work-related travel more broadly, and identify FIFO work as a specific and extreme case of work-related travel. We discuss the utility of considering FIFO work for theory development. We then focus more specifically on how FIFO work affects employee mental health and consider existing theories that address this topic, and theoretical gaps of these theories in capturing the key issues associated with FIFO work. Following on from this discussion, we report the method and findings of a systematic summary review of the psychological impact of fly-in-fly-out (FIFO) work. Finally, we draw on the literature review to develop a theoretical model linking FIFO work and mental health. This model serves as a guide for future investigations into the impact of FIFO work, and can be applied to other types of work-related travel.

Work-Related travel – an overview

Overall, work-related travel has been normalized for many workers (Gustafson, 2012), and working and living in separate locations has been identified as a common, modern-day phenomenon (Wood et al., 2015). In the research literature, the terms 'overnighters', or 'shuttlers', as well as multi-locality, have been used to describe those who engage in work-related travel. Multi-locality captures those who live and work in different places (Wood et al., 2015). It is useful to briefly reflect on these concepts that have been used to capture the various types of work-related travel, and to position FIFO work in relation to these broader concepts.

FIFO work is a specific form of work-related mobility that fits into what has been described as 'overnighting' or 'multi-locality'. Overnighting refers to the situation in which "people have to spend the night somewhere other than their home very frequently or for extended periods of time for job-related reasons" (Meil, 2014, p. 4). Specifically, overnighters are people who spend at least 60 nights away from their usual residence for work reasons (Rüger et al., 2011; 2016). For example, overnighting arises when the work is in remote areas, or there are high travel demands to multiple locations. While often a necessity due to spatial distances, in some cases, overnighting can arise because aspects of workers' personal life make this type of work arrangement practical and appealing (i.e., long-distance relationships; weekend commuters; preferences for family life; Rüger et al., 2016; Misan & Rudnik, 2015; Sibbel, 2010). A further concept that FIFO work broadly shares some features with is residential multi-locality. Residential multi-locality is a practice "which allows people to pursue their interests in different places either simultaneously or in alternating rhythms" (Wood et al., 2015; p.364). Put simply, multi-locality means that people live and work in different places, or simply reside in two locations. As observed by Wood et al. (2015) multi-local living is nowadays often expected from people and enabled by the trend of greater spatial mobility.

Work-related mobility requiring workers to stay away from home has an impact on workers and their families. In a recent literature review, Borowski et al. (2019) found eight articles that captured the effect of work-related travel on worker mental health. Specifically, Borowski et al. (2019) reported that strain and stress are common among frequent work travelers, as are sleep difficulties. Generally, this research literature shows that workers who spend a higher number of nights away from home show worse mental health (i.e., distress, anxiety, and depression; Rogers & Reilly, 2002; Rundle et al., 2018) and report higher work family conflict (Jensen, 2014; Mäkelä et al., 2014; Voydanoff, 2005). Montazer et al. (2020) identified that the linear effect of work travel duration on work-family conflict and psychological distress was evident for men, although not for women, whose overnight travel duration was not linked with work family conflict, or psychological distress. Using data from a large medical record data base (N = 18,328 employees), Rundle et al. (2018) reported that very high levels of business travel (i.e., > 21 nights per month) was associated with worse sleep, and at least mild anxiety and depressive symptoms. Reporting on insurance claims data, Striker et al. (2000) showed that the number of claims for psychological treatments rises with the number of business trips that a worker undertakes (see also Liese et al., 1997).

Beyond mental health, business travel has also been shown to affect workers lives more widely (Lück, & Schneider, 2010). For example, Rüger et al. (2011) report that women whose work involves job mobility are less often married and rarely have children, suggesting a possible impact of work-related travel on personal lives. Related to this, Rüger & Viry (2017) showed that the link between work related travel and fertility is dependent on the national context, suggesting that national policies related to family and work are important. It has been reported that highly mobile workers view their mobility as a positive choice, as well as an economic necessity, a constraint, or a burden (Viry, et al., 2015; Ravalet et al., 2017). However, between 13% to 33% of highly mobile workers from four European countries (Germany, Spain, France, Switzerland) perceive mobility negatively, and a noticeable number of them specifically report to 'never feel at home' (18% to 31%) and/or 'often feel tired' (38% to 58%; Viry et al., 2015).

The above summary illustrates the richness and diverse nature of this body of literature. However, as has been observed by Borowski et al. (2019), as well as Calderwood and Mitropoulus (2021), there is great variety in operationalization in travel for work research, and this research is distributed across different disciplines. Overall, the fractured nature of this body of research, as well as the varied nature of the way busines travel manifests in real life (i.e., duration, distances covered, travel mode, frequency), make it difficult to systematically identify theoretical aspects that are relevant. In this review, we argue that considering a case of business travel that is somewhat extreme, highly standardized, and has all the components of business travel is helpful for building a coherent model. FIFO work is such a case.

FIFO work as a specific case of spatial mobility: making the case for conceptual clarity

Clearly FIFO is a specific type of work that involves multi-local living and extensive travel. However, we identify that FIFO work shares several attributes that are generally applicable to work-related travel. The distance covered, commuting modality and commute duration, which in the case of FIFO work can be considered extreme, are key aspects of work-related travel that interact with work experiences, and are connected to worker satisfaction, strain and performance (Calderwood & Mitropoulos, 2021). Beyond the extreme commute, FIFO work shares the need to be located away from home over extended periods of time with other forms of work-related travel. In effect, these types of living arrangements require workers live two lives (Parkes et al., 2005). It is important to note that FIFO work, due to its controlled and structured nature, lends itself to clarifying and systematically reviewing aspects that may be applicable to other types of multi-locality or business travel. We argue here that FIFO work is particularly well suited to developing comprehensive insights on the effects of work-related travel because of its unique nature. Specifically, we identify three reasons that reinforce the usefulness of considering it.

First, FIFO work is an extreme form of work-related travel. In industries such as mining, oil and gas that rely on FIFO work, the work-related travel arrangements go beyond other forms of travel for work. Workers have very limited opportunity to remove themselves from the work context, usually follow regular rosters, are housed in employer accommodation, and are often away from home for long periods. Theory development and refinement can benefit from considering phenomena in more extreme contexts where processes may be more apparent and can help shed light on what people do in more ordinary contexts (Hällgren et al., 2018). As a result, a theoretical framework developed in the FIFO context can bring to light processes that apply to other forms of work-related travel but may be less apparent in these more ordinary arrangements (Stanko & Beckman, 2015). As Johns (2006) highlights, such extreme contexts can serve as "sharpeners of meaning".

Second, FIFO work represents a somewhat standardized case of work-related mobility in so far as most workers operating under this arrangement tend to be exposed to quite consistently similar conditions (as described above). This level of standardization allows a systematic view onto its various components that are less likely to be tainted by local variations or individual practices. Importantly, there is no clear agreement about how to operationalize work-related travel in the literature, with various studies focusing on different aspects, such as the destination, whether it involves international travel, or having to stay overnight (Borowski et al., 2019). Focusing on FIFO work allows some level of standardization and can help address these issues in the work-related travel literature more widely.

Third, unlike other forms of work-related travel, FIFO work is less likely to be diminished to the same extent as other types of work-related travel as a consequence of new work practices that emerged during Covid-19. During the pandemic, work-related travel has dramatically reduced and reliance on virtual work has increased (Abdullah et al., 2020; Herath & Herath, 2020); an impact that is likely to affect work-related travel beyond lockdowns (Caputo et al., 2021). While other types of work could adapt virtual work practices to a greater extent more permanently, FIFO work cannot be as easily replaced with virtual alternatives due to the nature of the work. The location of the work is bound by the location of the natural resources to which this work is connected and the physical nature of the work. Therefore, FIFO work is likely to persist in the future as a type of extreme work-related travel.

What is lacking from this body of research are clear insights into the psychological implications of FIFO work, especially how this extreme form of remote work can affect workers' mental health. Accordingly, it is important that we understand the impact of FIFO work on worker mental health, as well as what factors (e.g., work attributes) shape the effects of this practice. This review paper works towards addressing these two important issues and builds a theoretical model to guide research going forward.

Mental health and FIFO work

In this review, we follow the World Health Organization's (WHO) definition of mental health as a state of wellbeing in which the individual realizes their own potential, can cope with the normal stresses of life, can work productively and fruitfully, and is able to contribute to their community (World Health

Organization, 2013). Accordingly, we consider the whole spectrum of mental health ranging from mental ill-health to wellbeing, in line with the World Health Organization (2013) and the dual-factor model of mental health, proposed by Greenspoon and Saklofske (2001). These conceptualizations propose mental health as not only being characterized by the absence of mental illness, but to also include a sense of wellbeing. Accordingly, in this review we consider the aspects of mental ill- health and wellbeing as relevant to mental health. Mental ill-health indicators include burnout (i.e., a mental state characterized by emotional exhaustion, depersonalization and reduced personal accomplishment; Maslach, 1982), as well as anxiety and depression (Murray, Lopez, & World Health Organization, 1996; Restifo, Kashyap, Hooke, & Page, 2015; Newnham, Hooke & Page, 2010). Mental wellbeing can be represented based on three components (Lamers et al., 2011). First, emotional wellbeing, represents positive feelings of satisfaction and happiness, second, psychological wellbeing, indicates effective functioning of the individual (including aspects such as self-acceptance, personal growth, purpose in life, positive relations with others, autonomy and mastery) and third, social wellbeing, captures effective functioning in community life (including aspects such as social integration, social contribution, social coherence, social actualization and social acceptance).

Existing theories applicable to FIFO work: is what we have enough?

Existing theories or concepts that capture how work affects workers, such as work-family-conflict (Greenhaus & Beutell, 1985; Voydanoff, 2005), the job demands and resources model (Bakker & Demerouti, 2007, , 2017), and conservation of resources theory (Hobfoll, 1989; 2011), address some, but not all aspects of FIFO work.

First, work–family conflict literature, with its focus on tensions between the roles a person takes on at home and at work (Greenhaus & Beutell, 1985), is applicable to FIFO work. In FIFO work arrangements, while the worker is away, participation in the family role is made more difficult, if not impossible due to the worker often being physically removed from home. While this theory can help describe the conflict that workers experience it is somewhat limited by the processes that it describes as underlying the conflict. This limited applicability is rooted in the work-family conflict's focus on the tension between work and family roles as time based (i.e., time spent on one activity makes it harder to take part in another), strain based (i.e., strain experienced in one role will affect another role), or behavior based (i.e., behavior required in one role is incompatible with the other role; Greenhaus & Beutell, 1985). This list of conflict bases neglects the spatial separation that, in addition to temporal separation, is so central to FIFO work and has been identified as characterizing commutes more generally (Lyons & Chatterjee, 2008). In other words, the work family literature has predominantly focused on time-binds, rather than mobility binds (Hughes & Silver, 2020), yet both are key to understanding the impact of work-related travel, and specifically FIFO work, on workers.

Related to work family conflict are also the concepts of work and family segmentation and integration, which are also salient within the context of FIFO work. Segmentation represents "the degree to which aspects of each domain (such as thoughts, concerns, physical markers) are kept separate from one another—cognitively, physically, or behaviorally" (Kreiner, 2006; p.485). At the other end of the same continuum, integration describes "the merging and blending of various aspects of work and home" (Kreiner, 2006; p. 485). Individuals have segmentation and integration preferences (Nippert-Eng, 1996), and the extent to which each is present in the environment can be determined by workplaces (Hochschild, 1997). Importantly, the personal preferences and the workplace provision of a particular degree of segmentation and integration are connected to worker wellbeing (Kreiner, 2006). The spatial separation of FIFO work from home due to the vast distances covered by workers requires segmentation. Similarly, the rostered nature of FIFO work also dictates segmentation of work and home. Of note, while the workers are on roster, during their time off, they are forced to carry out activities related to their personal lives while residing in employer facilities (i.e., exercising, food consumption), which can promote integration. Boundary management strategies, commonly used to achieve the right balance between integration and segmentation by individuals and workplaces identified in the literature, such as full integration, segmentation, alternating, or dual centricity (Kossek & Lautsch, 2012) are however not applicable to FIFO workers. FIFO workers have limited say in the extent to which their work and home life is segmented or integrated given the controlled set up, the large distance between work and home and the need to be away from home for extended periods. These aspects of FIFO work limit worker's ability to engage in strategies to manage the boundaries between their work and personal lives.

A second relevant theoretical perspective is the job demands and resources model (Bakker & Demerouti, 2007; Karasek, 1979). This model is applicable to the FIFO context in that the theory captures the demands and resources that workers experience "on the job". Bakker and Demerouti (2007) described demands as attributes of a job that make it necessary for individuals to exert effort (e.g., high work pressure, an unfavorable physical environment, and emotionally demanding interactions with clients or colleagues). Job resources, as described by Bakker and Demerouti (2007) are attributes of jobs that help employees achieve goals, reduce demands, or stimulate personal growth (e.g., role clarity, or team support). These job demands and job resources are likely to affect FIFO workers to varying extents during their working hours. In fact, their availability may be very different to other work arrangements. However, the theory does not directly address how the extent of the impact of these demands and resources may play out differently in a context where employees are separated from their home lives and have sometimes only limited opportunity to remove themselves from the work context, compared to when they go home each day (Dorow & Jean, 2021). Additional demands that are specific to these types of work arrangements (e.g., long periods of travel, mentally transitioning between home and away time, being in remote locations) are also not captured in this model.

Third, the conservation of resources theory (Hobfoll, 1989; 2011) with its focus on resource loss and gain may be useful in describing the stark fluctuation between "at work" and "off work" that FIFO workers experience. A key principle of the theory is that individuals will strive to gain, retain and foster resources, and that this process requires them to invest resources (Hobfoll, 1989). Particularly relevant is the theory's consideration of people's behaviour when they are not faced with a stressor and how this behavior shapes the resources a person accumulates (Hobfoll, 1989). This aspect of the theory makes it particularly applicable to FIFO work as it may help to explain the ways in which prolonged exposure to work and extended periods of time-off interact to affect mental health in FIFO workers. Related to the conservation of resources theory are recovery and detachment from work. Recovery experiences (i.e., "experiences during leisure time that provide the opportunity to unwind from work"; Sonnentag & Bayer., 2005; p. 647) are important for workers to reduce strain reactions. Similarly, psychological detachment from work (i.e., "sense of being away from the work situation" p. 579; Etzion et al., 1998), has benefits for worker wellbeing. Both recovery and detachment are worthy of consideration in the FIFO work context as workers are somewhat limited in their ability to remove themselves from the workplace over extended periods of time, and their recovery activities can be somewhat limited. Further, the extent to which workers can detach through recovery activities will be affected due to any recovery activities being provided by the employer and occurring in a work-related context.

In sum, the key theories or conceptualizations that focus on how work affects workers' mental health provide some basis for theorizing that is applicable to FIFO workers. Importantly, however, they do not capture all aspects that are relevant to the specific experience of FIFO work. Accordingly, we propose that a specific theoretical model that captures the key aspects and processes involved in FIFO work is needed.

The Systematic Review: Our Approach

We conducted a systematic review of research that has considered the impact of FIFO work on worker mental health. By FIFO work, we mean work arrangements for which workers travel long distances to work locations and are required to stay at the location for extended periods (i.e., several days), in most cases in employer arranged accommodation. This includes both fly-in-fly-out (i.e., workers who fly to the worksite to cover substantial distances and overnight at the site) and drive-in-drive-out arrangements (i.e., workers who drive substantial distances to a worksite and overnight at the site).

In line with the dual-factor model of mental health (Greenspoon & Saklofske, 2001; see also World Health Organization, 2013), we include studies that cover any aspect of the spectrum of mental health, from mental ill-health (e.g., depression, anxiety, negative affect) to psychological wellbeing (e.g., life satisfaction, positive affect). The systematic review followed steps outlined by Liberati et al. (2009; see Figure 1) and included an electronic search using data bases with key words "fly-in-fly-out" OR FIFO OR "drive-in-drive-out" OR DIDO AND NOT "first-in, first-out" AND NOT Comput* (Source title – Publication name - Journal) and an additional hand search of material identified via other channels (subject matter experts, references identified in electronic search). These search terms were selected with the aim to be specific to identify research that is directly concerned with FIFO work. The search terms were used to search for research articles on Scopus, PubMed, Web of Science, Psycinfo, and Science Direct.

The search was conducted in April 2021 and returned 5610 papers. After one researcher assessed titles and removed duplicates, 223 abstracts were then independently analyzed by two researchers. Criteria for inclusion were that a paper needed to address FIFO work with a focus on work and workplaces, or workers, and involve data (that is, we excluded literature reviews and commentaries). We included peer reviewed journal papers, book chapters, theses, and other reports. Inter-rater agreement was above the recommended level of $\alpha = .80$ (see Krippendorff, 2004; p. 241) indicating that the two researchers were consistent in their judgments as to the items' relevance based on the abstracts' content (Krippendorff's $\alpha = .83$; $LL_{CI95\%}.72$ to $UL_{CI95\%}.94$). A final set of 64 full publications that both researchers had agreed on to be relevant were read by two researchers and their relevance was further screened to include those that addressed 1) the impact of FIFO work on workers' mental health directly, and/or 2) captured factors that may shape workers' mental health. Another inclusion criterion was that papers had to be written in English. Disagreements were discussed and resolved. A total of 37 papers were included in the final review.

Results

The relatively small number of studies employed a diverse range of methods including field studies, qualitative, mixed method, and quantitative approaches. Further, studies used a range of mental health indicators. The findings

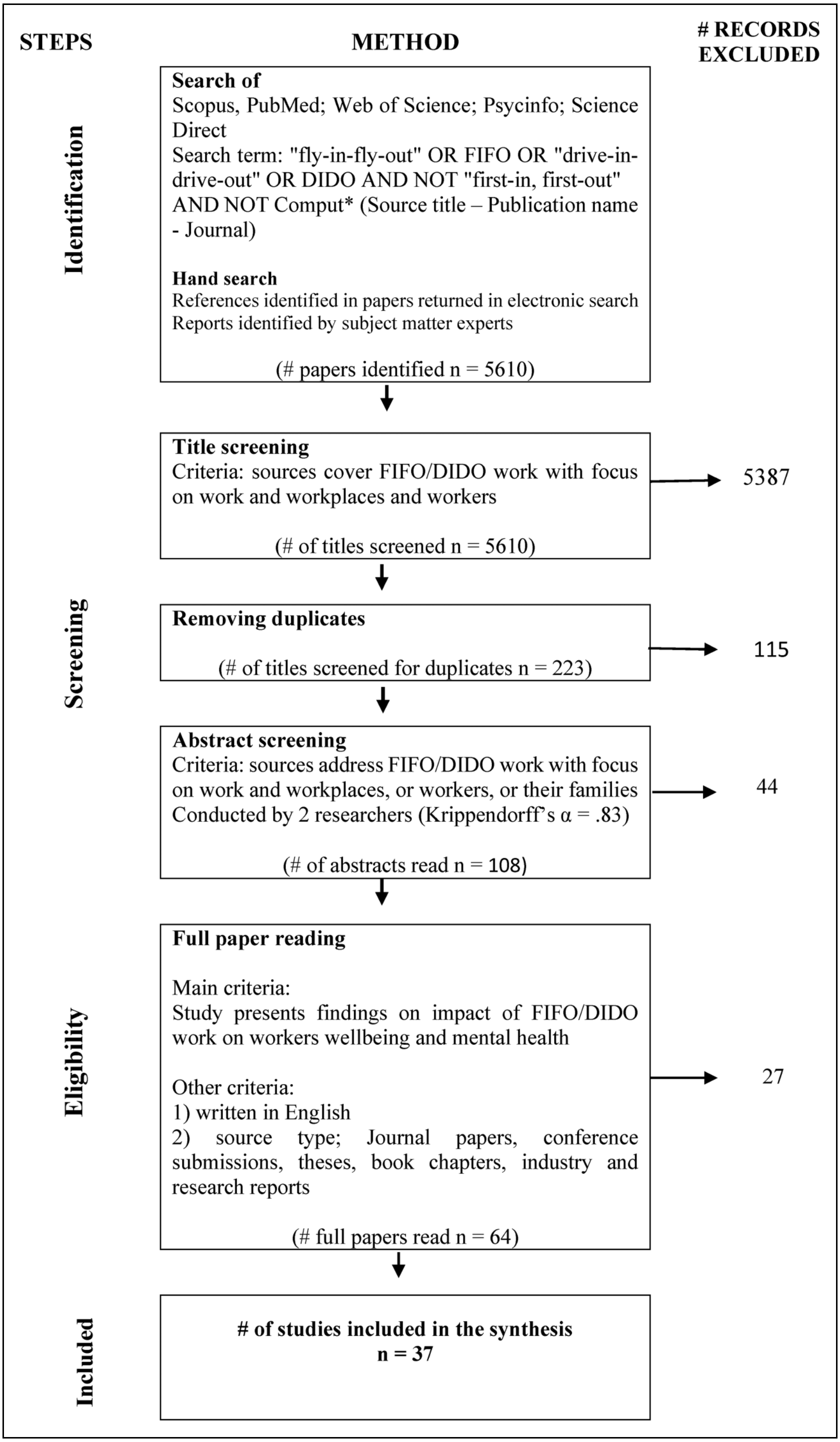

Figure 1. Overview of systematic literature review.

from each of the studies are useful in understanding the impact of FIFO work on the mental health of workers and contributing to the development of theory. However, the diversity within the studies (i.e., in terms of mental health conceptualization and operationalization, workplace aspects that were considered), and the numerous qualitative research included meant that a meta-analysis was judged not to be suited to sufficiently capture the research on FIFO work and mental health (Impellizzeri, & Bizzini, 2012). Accordingly, to best present the findings, we present a narrative summary overview of the findings (Ahn & Kang, 2018). The summary overview considers studies that have directly compared FIFO worker mental health with other workers and unpacks the factors that may be affecting worker mental health (see Table 1 in supplementary online material for full paper summary overview). We describe these findings first, and then discuss them in the context of the rigor of the overall research.

Levels of mental health amongst FIFO workers

Out of the identified studies, twelve reported a direct comparison of FIFO worker mental health and other workers. Across these studies, the majority reported FIFO workers to have worse mental health than other workers, or the general population, although findings are mixed. Seven studies report FIFO workers' mental health to be worse compared to workers in other similar forms of employment or population norms (Bowers et al., 2018, n = 1124; Lester et al., 2015, n = 23; Considine et al., 2017, n = 1457; Gilbert, 2019, n = 391; Henry, et al., 2013, n = 924; Parker et al., 2018, n = 3108; Sellenger & Oosthuizen, 2017, n = 105). Six of these studies used the K10 psychological distress measure (which assesses depression and anxiety; Kessler et al., 2002) and one the DASS-21 (Gilbert, 2019; which assesses depression, anxiety and stress; Lovibond & Lovibond, 1995). All these studies compared FIFO workers to general population samples or population norms.

In addition to the seven studies that found FIFO mental health to be worse based on a direct comparison with other populations or groups, fourteen studies also describe negative impacts of FIFO work on mental health in workers. One of these studies showed that stress levels of workers increased over the course of the roster while workers are on site, using saliva cortisol levels as an indicator in a longitudinal design (Korneeva & Simonova, 2020; n = 70). Complementing these findings, Carter (2008; n = 10) found FIFO workers to report depressive feelings in the time leading up to their return to work and during the first few days on site. Roster length and its impact on workers were also highlighted in a qualitative interview study by Martin (2020; n = 18) with FIFO workers at a site that has been associated with increased suicide levels in workers (between June 2017 and March 2018 five reported suicides, representing a rate of 167/100,000).

Miller et al. (2019; n = 751; 2020; n = 580) identified bullying on site as connected to suicide risk and depression in FIFO workers based on two independent survey data sets. The remaining eight of the studies reporting a negative effect of FIFO on worker mental health are in part focused on the impact of FIFO work on the social and family life of workers. This included the negative effects of missing family/ special occasions (Bradbury, 2011; n = 47; Gent, 2004; n = 132; Alroomi & Mohamed, 2021; n = 387) and limited opportunities for participating in hobbies or sports (Torkington et al., 2011; n = 11). The remainder of studies qualitatively describe increased risks of depression and anxiety, lack of privacy, social isolation and loneliness while workers are on site (Gilbert, 2019; n = 391; Pirotta, 2009; n = 20; Sibbel, 2010; n = 90; Dorow & Jean, 2021; n = 60). For example, based on interviews with FIFO workers, Dorow and Jean (2021) describe workers feeling trapped, routinized due to regimented schedules, andexperiencing monotony. As another example, Gilbert (2019) qualitatively reported factors affecting worker mental health such as the inflexibility of organizations to accommodate workers' needs, in rosters and point of hire, as well as the masculinity of FIFO workplaces contributing to mental health stigma, and a sense of having no clear path of exiting FIFO work.

In contrast to studies that report FIFO worker mental health to be worse than the mental health of other workers, four studies reported that FIFO work is associated with better mental health based on a direct comparison with those in other forms of employment, or in a mining job living residentially (Barclay et al., 2013, n = 286; Joyce et al., 2013, n = 380; Miller et al., 2019, n = 751; Velander, et al., 2010, n = 192). All but one of these studies used the DASS-21 questionnaire (Lovibond & Lovibond, 1995). The one study using a different measure was a study by Miller et al. (2019), which used Beck's Depression Inventory and the negative acts questionnaire. Moreover, all but Miller et al. (2019) compared FIFO workers to general population samples or population norms. Miller et al. (2019) used a comparison sample of residential workers in the resources sector.

Other studies that did not provide a direct comparison of FIFO workers' mental health and wellbeing with similar workers reported benefits of FIFO work. Four studies describe the benefits of FIFO work for worker mental health and aspects that are protective of worker mental health. For example, in qualitative interviews, workers themselves describe that they enjoy the lifestyle (Torkington et al., 2011; n = 11). Other qualitative interview studies specifically identify the benefits of the off-time that FIFO workers get during their rosters as it allows workers to spend quality time with friends and family and other activities that may be beneficial to their wellbeing (Bradbury, 2011; n = 47; Carter, 2008; n = 10), and provides a clear division of work and time off (Sutherland et al., 2017; n = 6). Carter (2008) also reports of workplace culture that supports a sense of camaraderie and mateship among employees while they are on site.

Finally, three studies identified no difference in the mental health of FIFO workers in comparison to similar forms of employment (Bradbury, 2011; n = 47; Clifford, 2009; n = 158; Gent, 2004; n = 132). Two of these studies also use the DASS-21 measure (Bradbury, 2011; Clifford, 2009). One study reported no difference on a general job satisfaction scale (Gent, 2004). Further, one descriptive study identified FIFO workers to have average job satisfaction and overall moderate levels of stress (Henry et al., 2013; n = 924). A study by Tuck et al. (2013; n = 157) reported the average levels of depression, stress, loneliness, and anxiety to be at moderate levels in their sample of FIFO workers.

Predictors of mental health amongst FIFO workers

We identified seven studies that statistically linked specific work and workplace attributes of FIFO workers with mental health outcomes. A survey study (Miller et al., 2020) identified social support as being positive for workers' mental health. In a survey study, Gilbert (2019) identified sense of community on site, as well as key work arrangement factors (including social aspects of the work environment, company support to attend to non-work issues, satisfaction with current work position, and quality of accommodation), as predictors of better mental health outcomes amongst FIFO workers. She also found perceived social support and sense of community, while lower for FIFO workers in comparison to other populations, to be strong predictors of subjective wellbeing.

Albrecht and Anglim (2018) surveyed FIFO workers on their work design perceptions, as well as their burnout and engagement over the course of a full onsite roster swing. Their results show that engagement and perceived supervisor support declined over the course of the roster, whereas emotional demand increased over time. They also found that perceived support from supervisors, colleagues or the organization was not associated with engagement. They reported a positive link between autonomy and engagement (at the day level) and that day level workload and emotional demands predicted emotional exhaustion.

Lester et al. (2015) studied distress, anxiety and depression (using the K10) in a small sample of 23 FIFO workers. Their results showed that FIFO workers on equal time rosters reported higher levels of distress than those on unequal time rosters (independent of whether these tend towards more days spent at work vs at home or the opposite). Related to rosters, Alroomi and Mohamed (2021) reported that the longer time workers had spent away working was linked with worse mental health (depression, anxiety, stress and fatigue). Tuck et al. (2013) found satisfaction with accommodation, recreation and social facilities, onsite support, contact with home, autonomy, onsite sleep quality, social relatedness to be associated with lower levels of depression, anxiety, and stress.

Finally, Parker et al. (2018) provide a comprehensive analysis into various aspects of FIFO work that affect workers' mental health. Using regression analyses, they reported FIFO specific work attributes such as isolation and loneliness on site, transitions between roster phases, rosters themselves (i.e., longer and uneven time rosters), accommodation characteristics (i.e., not having a permanent room), to be linked with poorer mental health. They also identified generic workplace attributes such as job insecurity, lack of support from team members and supervisors, high workload, and low of autonomy to be associated with worse mental health in workers.

Reflecting on the rigor and nature of the identified research

On balance, the review suggests that, on average, FIFO workers are likely to have worse mental health, especially in terms of anxiety and depression, than other workers. Importantly, there is variation in experiences, with various job, workplace and lifestyle attributes resulting in better or poorer mental health amongst some FIFO workers. Overall, we judge studies that report FIFO workers to have worse mental health to be more rigorous. Studies that identified worse mental health for FIFO workers tend to involve larger samples, suggesting that they are more representative of FIFO workers as a population. It is also worth noting that out of the seven studies reporting worse mental health for FIFO workers, four were published in peer reviewed journals. Similarly, all four of the studies reporting better mental health were published in peer reviewed journals. A starker difference in peer review frequency emerges between the studies reporting impacts of FIFO work on mental health. Eight of the studies reporting negative impacts of FIFO work on mental health were peer reviewed. Out of the studies reporting positive impacts of FIFO work on mental health, only two were published in peer reviewed journals. Studies reporting no differences were reported in an Honors thesis and two PhD theses. Overall, the sample sizes, use of standardized measures, and level of peer review reinforces the higher rigor of studies reporting negative impacts and worse mental health in FIFO workers.

We identify four specific issues that are important to note about the research on FIFO work and mental health. First, much of the research we identified is exploratory, with many studies being qualitative rather than quantitative, with small sample sizes. Of note, in some of the qualitative studies, samples are small (e.g., Carter, 2008; $n = 10$; Perring et al., 2014; $n = 7$; Sutherland et al.; $n = 6$) and did not reach the 10–13 sample size criterion for saturation suggested by Francis et al. (2010) or did not report the sample size (Haslam McKenzie & Hoath, 2016). With regards to methodological rigor, most studies employ validated measures, however as an example, the study by Alroomi and Mohamed (2021) combined scores from selected items of four separately validated and developed scales

(DASS-21, HADS, HSCL-25, CES-D) measuring different concepts into one overall 'mental health score'. Exploratory work has its place in the research process and is very useful in identifying specific research questions. However, the nature of the research, particularly the small sample sizes in some of the studies, may have contributed to lack of rigor.

Second, comparison studies reporting differing findings of the level of FIFO mental health systematically varied in the type of mental health measure used. Studies finding more positive mental health or no difference in FIFO workers compared to others tend to use the DASS-21 measure, whereas those that report worse mental health in FIFO workers predominantly employ the K10. Differences in these measures might help explain the differences in findings. The DASS-21 measures depression, anxiety and stress (Lovibond & Lovibond, 1995), whereas the K10 measures anxiety and depression only (Andrews & Slade, 2001). Further, the DASS-21 asks responders about the previous 10 days, whereas the K10 enquires about the previous 30 days. Finally, the DASS-21 measures anxiety with a focus on the physical symptoms of anxiety, whereas the K10 does not cover physical symptoms.

Third, causality is not clearly addressed in the research papers we identified. Out of the papers only two adopted a longitudinal research method (Albrecht & Anglim, 2018; Parker et al., 2018), focusing on the fluctuation in mental health over time, or within person variation, in the perceptions of workplace attributes over time (i.e., supervisor support). With many papers being descriptive or exploratory, clear conclusions on causality and regarding unobserved heterogeneity cannot be drawn.

Fourth, in addition to these methodological issues, we found that a coherent theoretical model that can systematically guide research in this area and capture the central aspect of FIFO work, is absent from this literature. In many cases papers are lacking theoretical grounding (see also Albrecht & Anglim (2018) for a similar observation). Specifically, our review indicates that most studies do not explicitly refer to a theory or a theoretically derived conceptual basis (see Table 1; this was the case for 28 of the research papers that we identified). Of the studies that refer to a theoretical or conceptual framework, five referred to work family conflict concepts, including segmentation and boundaries between work and family life (Bailey-Kruger, 2012; Bradbury, 2011; Carter, 2008; Gent, 2004; Sibbel, 2010), four referred to the jobs demands and resources model (Albrecht & Anglim, 2018; Carter, 2008; McTernan et al., 2016; Parker et al., 2018), and one referred to social support literature and sense of community theory (Gilbert, 2019). The inclusion of the jobs demands and resources model (Bakker & Demerouti, 2007) as well as work family conflict literature (Greenhaus & Beutell, 1985) is aligned to the theories that we identified above as related to FIFO work. However, it should be noted that the conservation of resources theory (Hobfoll, 1989), although identified above as applicable to FIFO work, is not currently referred to in the FIFO work literature. This is possibly due to the mostly static view that the literature has so far adopted to FIFO work's impact on workers that does not take fluctuation in resource diminishment and recuperation into account.

In summary, it seems a fair assessment that FIFO work research to date is somewhat unsystematic, mostly descriptive, and overall lacking coherent theory. The lack of theory may be due to the predominantly descriptive and exploratory nature of the research. However, it also needs to be considered that there is no theory available in the literature that adequately captures FIFO work. It is to this we now turn.

Blending and fracturing of life: a theoretical model to capture FIFO worker experiences in future research

Two attributes of FIFO work are central to the theorizing of its impact on workers' mental health: the blending of personal life and work while workers are on site, and the simultaneous fracturing of work and home life. This contrasting setup occurs due to the rostered nature of this work arrangement, where workers are away for prolonged times and then come back home for an extended period (e.g., Storey, 2001). Importantly these two aspects of FIFO work have not been brought together in one theoretical model yet doing so is critical for our understanding of FIFO work's impact on worker mental health. In fact, it has been pointed out that FIFO workers live in effect "two lives" between their work and home life (Parkes et al., 2005), and that the managing of multiple roles between work and home life is one of the key challenges of FIFO work (Gardner et al., 2018; Gilbert, 2019).

Conceptually, it is important to consider that the two phases that FIFO workers constantly cycle through are quite different and reveal a pattern of extremes. First, workers' limited ability to separate their personal time after work from the work context represents a paradoxical contrast to the simultaneous separation from their actual home environment. Second, on site, workers may have limited control over their decision-making, and many of their actions are prescribed, scheduled and organized for them (Dorow & Jean, 2021; Gilbert, 2019). This contrasts with their home life, where schedules and regular plans do not exist, or are developed in the workers' absence. They might therefore struggle to fit into a family and social life where a routine has developed that may not include them and social activities and hobbies may be difficult to sustain (Gilbert, 2019; Parker et al., 2018). Other differences that these two contexts may present include their location (i.e., remote locations (at work) vs urban locations (at home)), and differences in the pace of life (i.e., a sprint (at work) vs a sudden pause (at home; Gilbert, 2019). Consequently, there is not a happy modicum for these workers where they get a bit of both sides of personal and work life.

The fracturing and blending described in our model share some conceptual overlap with segmentation and integration, described by Kreiner (2006). However, we propose that fracturing and blending represent more profound forms of what has been described as segmentation and integration. Segmentation and integration occur at daily, micro level, often with a lot of worker discretion to adapt the level of integration and segmentation to their personal preferences (Kreiner, 2006). In contrast fracturing and blending signify two aspects of FIFO work that mostly occur over weekly periods and are dictated by employer arrangements. Fracturing and blending are rooted in different work-related factors, and ultimately represent different phenomena with different processes involved compared to segmentation and integration. As is described below, FIFO workers are limited in their ability to determine the extent of fracturing and blending that they experience. Furthermore, FIFO work arrangements prescribe physical blending and fracturing alongside psychological blending and fracturing, representing a unique feature. Of note, these two are not always in step. For example, a worker may be located at the FIFO campsite away from home (physical fracturing), but they may be calling their family and thinking about them (psychological blending). Existing work family conceptualizations around boundary segmentation and integration related to work family conflict (e.g., Voydanoff, 2005; Kossek et al., 2005) do not account for the concurrent experience of both extremes of blending and fracturing in FIFO work and do not account for extreme fluctuation as is experienced by FIFO workers. Further, as noted above, boundary management strategies by individuals and workplaces identified in the literature, are not options available for FIFO workers. Collectively, these unique aspects illustrate the distinct nature of fracturing and blending from segmentation and integration.

Proposition 1: FIFO work is characterized by the simultaneous blending and fracturing of work and personal life that workers experience.

FIFO work arrangements create a work design that requires workers to integrate and bridge paradoxical extremes. Cycling through these extremes and constantly adapting to change (e.g., in location, context, pace, routine) will likely pose meta-demands on workers. These meta-demands are not directly evident in the work design on site or the workers' lives while at home per se. Consequently, they are not represented in existing theories, such as the jobs-demands and resources theory (Bakker & Demerouti, 2007). These demands exist between two states that the workers need to constantly manage when psychologically transitioning from one extreme to another. As part of that process, they need to manage their mental health, and put effort into integrating their social and family lives (Taylor & Simmonds, 2009). The demands that come from doing so may involve reflecting and planning, coping, but also social interactions that the workers engage in to maintain life in two spheres. Consequently, the continuous blending and fracturing nature of FIFO work likely contributes to its impact on worker mental health. Future research into the ways in which workers maneuver through these two ends of extremes, at times experienced simultaneously, is key.

> ***Proposition 2***: The continuous blending and fracturing of work and personal life that FIFO workers experience negatively affects workers' mental health.

In what follows, we present a model that captures these two core components of FIFO work and by extension other work-related travel (Figure 2).

Blending of work and personal life

Based on the concept of liminality, Dorow and Jean (2021) suggested that, during their time on site, FIFO workers experience a blending of work and life. There are some specific aspects of life away from home for FIFO workers and by extension other forms of work-related

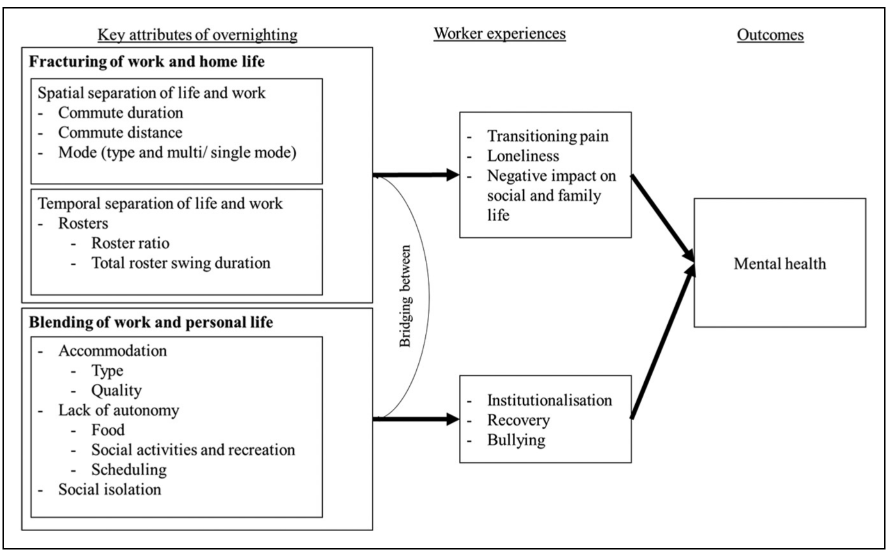

Figure 2. Model of blending and fracturing of work and life experiences in FIFO work.

travel that may contribute to the blending of life and work. First, lack of autonomy shapes workers' experiences while on site and at the accommodation. Autonomy can be paraphrased as freedom of choice (Kühler & Jelinek, 2013) and its absence can manifest in various restrictions for FIFO workers. Lack of autonomy experienced by FIFO workers may manifest in various ways, starting with lack of choice around food, accommodation, and being in charge of one's own schedule and activities (Dorow & Jean, 2021; Johnsen & Sørensen, 2015). In particular, around accommodation, quality can vary, and workers may have varying levels of privacy as they may be sharing a room (often with a worker who works the opposite shift; Haslam McKenzie & Hoath, 2016; Sibbel et al., 2016). Of note the limitations to autonomy experienced by FIFO workers extend beyond issues that are commonly focused on in the job demands and resource model (Bakker & Demerouti. 2007), which considers autonomy specific to the work, or tasks. The theory specifies, method, scheduling, and decision-making autonomy (Breaugh, 1985) with a focus on work content directly. However, the limitations to autonomy experienced by FIFO workers reach far beyond decisions as to how they carry out their work. The autonomy limitation experiences by these workers infringe on their personal lives and personal choices much more than is traditionally captured by work related autonomy.

Next, the workers' ability to actively leisure craft may be limited by the facilities and options available to them. Leisure crafting is the "pursuit of leisure activities targeted at goal setting, human connection, learning and personal development" (Petrou & Bakker, 2016; p. 508). After work hours, FIFO workers (and people who travel for work more generally) are limited to the resources and recovery options that their employers make available to them. Because of the routinized scheduling of activities outside of work hours, Dorow and Jean (2021) identified that workers may often experience monotony and boredom. For FIFO workers, sporting facilities, such as a pool, or gym may be available (Parker et al., 2018). On FIFO sites, the dining mess is often a place where people come together socially to eat and in some cases to drink (if permitted) and workers identify social facilities and "village lifestyle factors" as important for their wellbeing (Perring et al., 2014; Sibbel, et al., 2016). While these recreational facilities and activities exist to simulate personal recreation, they cannot address the fact that workers are engaging with them while located on employer operated grounds, with work colleagues, so that actual detachment from work is impossible. The meshing this presents goes beyond what has been described in the work family literature as integration (e.g., Kreiner, 2006). The extent to which these activities are mixed up represent a merging as the personal activities that a worker can undertake while on site (e.g., eating, exercising) are in essence no longer personal, when they involve employer provided equipment and occur in an employer controlled environment.

Further, social isolation and lack of relatedness while away can also contribute to the experience of blending of life and work (Tuck et al., 2013). FIFO workers' social isolation on site has a critical impact on their mental health (Sellenger & Oozthuizen, 2017). The issue of social isolation illustrates how the blending of work and life on site makes true connection and social contact hard to come by. Contact and friendships may be fleeting and often limited to colleagues, which clearly evidences how overcoming social isolation on site is almost impossible without the blending of life and work (Saxinger, 2021).

> ***Proposition 3***: Blending of work and personal life on site/ while away in FIFO work occurs due to lack of autonomy, limited opportunities to leisure craft, and social isolation.

Mechanisms linking blending of work and personal life while away/ on site with mental health

We identify three key impacts of the blending of work and personal life on workers that can contribute to mental health issues in FIFO workers. First, workers may experience institutionalization when their autonomy is highly limited due to prescribed schedules and limited choices (Dorow & Jean, 2021; Gilbert et al., in prep). This institutionalization may affect their ability to make decisions for themselves within and outside of the blended work and personal life context (Geppert & Pastuh, 2017).

Second, recovery and detachment from work that workers can realistically achieve in blended contexts will be limited. Recovery as described above captures processes that allow individuals to unwind and restore following work (Sonnentag & Fritz, 2007). Recovery depends on the ability of the worker to remove themselves from the work environment not only physically but also psychologically (Sonnentag & Bayer, 2005), as well as mastery experiences and relaxation (Sonnentag & Fritz, 2007). Similarly, detachment both physically and psychologically will be limited for FIFO while they are on site due to the blending that they are exposed to.

Third, bullying is likely related to the blending of work and life while FIFO workers are away from home. We include bullying here as this was identified by a number of studies on FIFO work arrangements as an issue that affects worker mental health negatively (Sellenger & Oosthuizen, 2017; Miller et al., 2019; 2020). Bullying constitutes the repeated exposure to behaviour that is experienced to be intimidating, humiliating, or frightening (Leymann, 1996). Bullying may be more likely and its impact possibly more severe for FIFO workers as they have limited opportunity to remove themselves from the context in which it occurs. These contextual attributes may further exacerbate the impact of bullying on workers.

Proposition 4: Blending of work and personal life on site/ while away contributes to institutionalization, difficulty to recover and detach from work, and bullying in FIFO workers.

Fracturing of work and home life

The second key attribute that characterizes how FIFO work and by extension other types of work-related travel affects workers is the extent to which it fractures life. Because of the rostered work, workers have extended and intensive periods in which their life is very much focused on work, followed by an extended time off at home. Between work and home life, a clear spatial and temporal separation exists for FIFO workers. We acknowledge that such a clear separation may have benefits for workers and may present an opportunity for quality time off during which workers can truly switch off from work (Bradbury, 2011; Carter 2008; Sutherland et al., 2017). This is indeed the case. However, we suggest that on balance the fracturing of work and home life is more likely to be detrimental. This effect may be related to the extent to which FIFO work poses work family conflict. Research on the impact of FIFO work on family life shows it to be linked with greater perceptions of family dysfunction compared to other families (Kaczmarek & Sibbel, 2008), and less parental presence and family connectedness compared to non-FIFO families (Lester et al., 2015). As previously mentioned, we posit that the fracturing that FIFO workers experiences goes beyond what has been described in the work family literature as segmentation (Kreiner, 2006).

We identify two specific aspects as relevant to workers' experiences of fracturing, namely the commute (i.e., capturing spatial fracturing) and rosters (i.e., capturing temporal fracturing). There are good conceptual reasons to suggest that the commute is critical to our understanding of the ways in which FIFO work affects workers (Clifford, 2009). First, the distances that FIFO workers travel are, as pointed out above, very large (e.g., Storey, 2001). Research on daily commutes shows that longer commutes are linked to higher levels of worker strain and worse mental health (Clark et al., 2020). Moreover Chatterjee et al. (2020) showed that mood is negatively impacted during the commute itself. We posit that it is not unreasonable to suggest that FIFO workers who are commuting these extremely long distances may be similarly affected by their commuting experience. Second, the commute also presents an actual and psychological separation between the workers' work and home lives. This separation manifests in the duration of the commute and the spatial distance that workers cover. It may make returning home in case of family emergencies or other instances more difficult. Therefore, the commute in terms of duration, distance and mode is key to understanding the fracturing of life experienced by FIFO workers.

Second, we propose the rosters that workers cycle through as central to the fracturing experiences of workers. The full roster captures the time that workers spend on site working and the period of time-off that follows. There is a high variability in roster lengths and attributes across industries and countries (Parkes, 2015). Moreover, research findings are mixed for the various types of rosters. For example, Parker et al. (2018) showed that workers on rosters with a higher ratio of time spent on site, than off work (i.e., 4 weeks on/1 week off, 3 weeks on/1 week off and 2weeks on/1 week off) had higher levels of psychological distress than those on other, more balanced rosters (i.e., even-time rosters 2 weeks on/ 2 weeks off, 8 days on/ 6 days off). Similarly, research by Martin (2020) and Devine et al. (2008) point to the detrimental effects of rosters in which workers spend long periods on site, compared to off site. In contrast, findings by Bowers et al. (2018) report that workers on a 4 weeks on/ 1 week off rosters had better mental health than those on a 2 weeks on and 1 week off roster. Other studies show no link between roster type and worker mental health (Sibbel et al., 2016).

Given these mixed findings, it appears that roster patterns on their own may not directly explain how FIFO work affects workers. This makes sense when seen in the context of established work design theories (e.g., Bakker & Demerouti, 2007) which would suggest that for example a worker who has great support while on site and good levels of autonomy will be much differently affected by the time that they spend on site, compared to someone with a less advantageous work design. To address this issue, we propose to understand the impact of rosters it is necessary to consider the extent to which they fracture workers' lives.

Further, the impact of roster types on workers' fracturing of life can be softened by experiences of the workers. First, we identify flexibility from employers as a buffer that may support workers in connecting between work and home life. Importantly, FIFO work generally allows little flexibility with regards to flextime (i.e., flexibility about when work is done) or flexplace (i.e., flexibility about where work is done; Shockley & Allen, 2007). In the FIFO work context, however, there is some discretion from the employers' side that allows them to provide some flexibility. Flexibility in this context has been defined as "the degree of flexibility that FIFO workers have, such as the option of job sharing, time off for important events or requests for different rosters" (Parker et al., 2018, p. 103). This flexibility is key to workers bridging the facture that exists between their work and home lives when this is acute (e.g., in the case of family emergencies) and on an ongoing basis when a roster can be offered that suits the workers' lives. Second, connecting with home while workers are on site can bridge the fracture between home and work lives (Gardner et al., 2018; Gilbert, 2019; Tuck et al., 2013; Taylor & Simmonds, 2009). The ability to connect with home can be challenged by poor communication facilities (i.e., the quality of the internet or phone connection, if any in very remote locations), but also an oftentimes long day that makes fitting a call in difficult (Parker et al., 2018). FIFO workers often work 12 h shift from 6am to 6pm and in many cases, this time is preceded or followed by a commute from work site to camp. Depending on restrictions (i.e., do the workers need to change first) and the scheduling of dinner, workers, in particular those with young children, may struggle to stay connected with home (Parker et al., 2018). Research has shown that FIFO work hours and shift length are linked with child behavioral problems (Dittman et al., 2016) and the impact of such long hours on the FIFO workers ability to bridge the fracturing between work and home life may be key.

> ***Proposition 5***: Fracturing of work and home life in FIFO work occurs due to lengthy commutes (i.e., capturing spatial fracturing) and extended rosters (i.e., capturing temporal fracturing).

Mechanisms linking fracturing of work and home life with mental health

We identify three immediate impacts of the fracturing between work and home life in FIFO work that have potential implications for mental health. First, workers may face transition pains, i.e., psychological challenges […] while transitioning between site and home (Parker et al., 2018; p. 102). It is important that the transitioning pains we are including here are not about the commute, or physical experiences, but about the psychological demands involved in maneuvering between work and home lives when these two aspects are so fractured. Role transitions have generally been described as involving psychological adjustment from one role to the next (Burr, 1972). Due to their prolonged absences both from work and home, workers need to find their place again every time they return to site or in particular home, where life may have moved on while they have been away. In addition, it is often hard for workers transitioning back onto site, which can be emotionally stressful (Parker et al., 2018). Of note these types of transitions involved in FIFO work are different to what has been described by Ashforth. (2000) who distinguish micro transitions (i.e., daily commutes) and macro transitions (i.e., permanent relocations).

A second immediate impact of the fracturing of life experienced by FIFO workers is loneliness. Loneliness is a subjective experience, which results from insufficient social relations and is distressing for the person who experiences it (Franklin et al., 2018; Perlman & Peplau, 1981). It has a reciprocal relationship with subjective wellbeing in the wider public (van der Weele et al., 2012). The fracturing of their lives makes it harder for FIFO workers to stay in touch and creates a sense of disconnection from others. In fact, FIFO workers themselves report to experience loneliness (Barclay et al., 2013; Watts, 2004).

Finally, the fracturing of FIFO workers' work and home lives is also likely to be associated with work-family-conflict. Work–family conflict captures the tensions between the roles a person takes on at home and at work (Greenhaus & Beutell, 1985). Indeed, FIFO work has been reported to create such tensions (Dittman et al., 2016). We propose that the fracturing of work and home lives that FIFO workers experience is key to understanding its impact on work-family-conflict. The fracturing that occurs is both due to spatial and temporal separation of home and work lives. FIFO workers and their family are facing both temporal and spatial separation and subsequently role conflict related to both these separations. This situation may in fact mean that work family conflict is enhanced in FIFO work arrangements (Gilbert, 2019).

> ***Proposition 6***: Fracturing of work and home life contributes to transition pains, loneliness, and work–family conflict in FIFO workers.

The FIFO worker – personal dispositions and FIFO work experiences

It is important to acknowledge that different people may react differently to FIFO work. The extent to which a FIFO worker's mental health is affected may depend on several personal attributes, some of which we reflect on here. First, preferences akin to segmentation and integration preferences (i.e., the extent which individuals prefer their work and personal lives to overlap; Kreiner, 2006) may also apply in relation to fracturing and blending. When seen in the context of FIFO work, it may for example be that someone who prefers their work and home lives to be fractured will thrive while at home and clearly being removed from their work. These individuals however will be more negatively affected while they are on site during their time off when the blending of personal and professional lives occurs. Someone who prefers blending in their work and personal life, would likely benefit from the blending of work and personal life at camp, while struggling more with being away from family while at work.

Next, personal resilience may also support FIFO workers in adjusting to the challenges of FIFO work. Resilience describes the "positive psychological capacity to rebound, to 'bounce back' from adversity, uncertainty, conflict, failure, or even positive change, progress and increased responsibility" (Luthans, 2002; p. 702). A FIFO worker's ability to take things in their stride, be on their own, or bounce back from adversity (Luthans et al., 2007) may assist them in living with the extremes that FIFO work presents.

Finally, the type of commitment that workers experience around FIFO work may also be important. In the FIFO work literature, the term "golden handcuffs" (Palmer, 2014) has been used to refer to a concept akin to continuance commitment. In essence, the golden handcuffs describe a phenomenon in which people are attracted to FIFO work because it pays very well. Overtime some individuals may unintentionally lock themselves into FIFO work, due to overspending on houses, toys, and other expenses (Palmer, 2014). Similar to continuance commitment, the "golden handcuffs" mean that someone remains attached to an organization or occupation, because they need to (Meyer et al., 1990). It is likely that those FIFO workers who have locked themselves into FIFO work due to overspending will struggle more with the overall lifestyle than those who operate with the option of changing to a different employment form if they wish to do so.

Limitations and future research

This paper provides a systematic overview of the research literature into FIFO work and mental health. As stated above, we opted for a systematic review to capture a methodologically diverse body of literature that has considered a wide range of workplace attributes as well as mental health indicators. This approach aligns with guidance in the literature (Impellizzeri, & Bizzini, 2012). However, we acknowledge that a meta-analytical approach has benefits beyond the systematic description of a body of research. Specifically, a meta-analysis would have had the advantage of identifying the overall statistical effect of FIFO work on worker mental health (Egger et al., 1997). Accordingly, in the future, a meta-analytical summary of the research finding on FIFO work and mental health will be warranted.

Building on the theoretical model developed in this paper, we proposed that future research can conduct a direct test of the model components in FIFO workers. Doing so will identify the applicability and relevance of its components to FIFO work. Specifically, we further identify that longitudinal studies are needed that can track the impact of blending and fracturing over time. Importantly, such longitudinal studies can capture how cycling through these two extremes may affect workers and how they manage this process. Beyond the FIFO context, the model may also proof useful to other contexts that involve work related travel and having to stay away from home due to work. The extent to which this is the case can be explored in future research.

Conclusion

This review illustrates the ways in which FIFO work affects workers' mental health.

Specifically, it identifies the simultaneous blending and fracturing of work and personal lives as central aspects that are necessitated by this type of work arrangement. Of note, the research literature in this area was found to not consistently adopt an existing theoretical model or approach. Importantly the here developed model of blending and fracturing brings together the various principles and processes identified in this somewhat scattered literature and provides a research model that more coherently captures FIFO work and its impacts on workers. This model can help build a coherent and systematic body of research on FIFO workers mental health going forward. While developed in reference to FIFO work specifically, we propose that the principles of blending and fracturing of home and work life may also be applicable to other types of work arrangements that involve work-related travel to varying degrees.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the Western Australian Mental Health Commission (grant number MHC 17/2944).

ORCID iD

Laura S. Fruhen https://orcid.org/0000-0002-9599-3468

Supplemental material

Supplemental material for this article is available online.

Note

1. According to Australian Bureau of Statistics Census results from 2016 Western Australia's Labour Force Population was 1,255,706 see here: https://www.abs.gov.au/ausstats/abs@.nsf/mediareleasesbyReleaseDate/B9F36A1DE12770E7CA2581BF001F7427?OpenDocument

References

Abdullah, M., Dias, C., Muley, D. & Shahin, M. (2020). Exploring the impacts of COVID-19 on travel behavior and mode preferences. *Transportation Research Interdisciplinary Perspectives*, *8*, 100255. https://doi.org/10.1016/j.trip.2020.100255

Ahn, E. & Kang, H. (2018). Introduction to systematic review and meta-analysis. *Korean Journal of Anesthesiology*, *71*(2), 103–112. https://doi.org/10.4097/kjae.2018.71.2.103

*Albrecht, S. L. & Anglim, J. (2018). Employee engagement and emotional exhaustion of fly-in-fly-out workers: A diary study. *Australian Journal of Psychology*, *70*(1), 66–75. https://doi.org/10.1111/ajpy.12155

*Alroomi, A. S., & Mohamed, S. (2021). Predictors of mental health and fatigue among isolated oil and gas workers. Predictors of mental health and fatigue among isolated oil and gas workers, *Safety and Reliability*, *18*(21), 11700. https://doi.org/10.3390/ijerph182111700

Andrews, G. & Slade, T. (2001). Interpreting scores on the Kessler psychological distress scale (K10). *Australian and New Zealand Journal of Public Health*, *25*(6), 494–497. https://doi.org/10.1111/j.1467-842X.2001.tb00310.x

Ashforth, B. (2000). *Role transitions in organizational life: An identity-based perspective*. Routledge.

Australian Bureau of Statistics. (2016). Suicide in Australia. Retrieved from http://www.abs.gov.au/ausstats/abs@.nsf/Lookup/by%20Subject/3303.0~2016~Main%20Features~Intentional%20self-harm:%20key%20characteristics~7

*Bailey-Kruger, A. (2012). The psychological well-being of women operating mining machinery in a fly-in fly-out capacity. Master's thesis, Murdoch University, Western Australia, Australia. Retrieved from http://ro.ecu.edu.au/theses/1682

Bakker, A. B. & Demerouti, E. (2007). The job demands-resources model: State of the art. *Journal of Managerial Psychology*, *22*(3), 309–328. https://doi.org/10.1108/02683940710733115

Bakker, A. B. & Demerouti, E. (2017). Job demands–resources theory: Taking stock and looking forward. *Journal of Occupational Health Psychology*, *22*(3), 273–285. https://psycnet.apa.org/doi/10.1037/ocp0000056

*Barclay, M. A., Harris, J., Everingham, J., Kirsch, P., Arend, S., Shi, S. & Kim, J. (2013). Factors linked to the well-being of fly-in-fly-out (FIFO) workers. *Research Report*, CSRM and MISHC, Sustainable Minerals Institute, Brisbane, Australia: University of Queensland.

Borowski, S. C., Naar, J. J., Zvonkovic, A. M. & Swenson, A. R. (2019). A literature review of overnight work travel within individual, family and social contexts. *Community, Work & Family*, *22*(3), 357–383. https://doi.org/10.1080/13668803.2018.1463970

*Bowers, J., Lo, J., Miller, P., Mawren, D., & Jones, B. (2018). Psychological distress in remote mining and construction workers in Australia. *The Medical Journal of Australia*, *208*, 391–397. https://doi.org/10.5694/mja17.00950

*Bradbury, G. S. (2011). Children and the fly-in/fly-out lifestyle: Employment-related paternal absence and the implications for children. PhD thesis, Curtin University, Western Australia, Australia. Retrieved from https://espace.curtin.edu.au/

Breaugh, J. A. (1985). The measurement of work autonomy. *Human Relations*, *38*(6), 551–570. https://doi.org/10.1177/001872678503800604

Burr, W. R. (1972). Role transitions: A reformulation of theory. *Journal of Marriage and the Family*, *34*(3), 407–416. https://doi.org/10.2307/350436

Calderwood, C. & Mitropoulos, T. (2021). Commuting spillover: A systematic review and agenda for research. *Journal of Organizational Behavior*, *42*(2), 162–187. https://doi.org/10.1002/job.2462

Caputo, P., Jackson, A. J., Murali, R., Rauch, M. & Terry, B. (2021, August 02). *Return to a world transformed How the pandemic is reshaping corporate travel*. Retrieved from https://www2.deloitte.com/us/en/insights/focus/transportation/future-of-business-travel-post-covid.html

*Carter, T. (2008). An exploration of Generation Y's experiences of offshore Fly-in/Fly-out (FIFO) employment. Honours thesis, Edith Cowan University, Western Australia, Australia. Retrieved from http://ro.ecu.edu.au/theses_hons/1166/

Chatterjee, K., Chng, S., Clark, B., Davis, A., De Vos, J. & Ettema, D., … & Reardon, L. (2020). Commuting and wellbeing: A critical overview of the literature with implications for policy and future research. *Transport Reviews*, *40*(1), 5–34. https://doi.org/10.1080/01441647.2019.1649317

Clark, B., Chatterjee, K. & Martin, A., & Davis, A. (*2020*). How commuting affects subjective well-being. *Transportation*, *47*, 2777–2805. https://doi.org/10.1007/s11116-019-09983-9

*Clifford, S. (2009). The effects of fly-in/fly-out commute arrangements and extended working hours on the stress, lifestyle, relationship and health characteristics of Western Australian mining employees and their partners. PhD thesis, The University of Western Australia, Western Australia, Australia. Retrieved from http://research-repository.uwa.edu.au/en/

*Considine, R., Tynan, R., James, C., Wiggers, J., Lewin, T., Inder, K., Perkins, D., Handley, T., & Kelly, B. (2017). The contribution of individual, social and work characteristics to employee mental health in a coal mining industry population. *PLoS ONE*, *12*(1), e0168445. https://doi.org/10.1371/journal.pone.0168445

Costa, S. D. Silva, A. C & Hui, V. (2006). What are the opportunities and challenges for women in fly-in-fly-out (FIFO) mining in Canada? An exploratory study. The CIM Vancouver 2006 conference and exhibition: creating value with values, Vancouver, BC (Canada), 14–17 May 2006.

*Devine, S. G., Muller, R., Carter, A. (2008). Using the framework for health promotion action to address staff perceptions of occupational health and safety at a fly-in/fly-out mine in north-west queensland. *Health Promotion Journal of Australia*, *19*(3), 196–202. https://doi.org/10.1071/HE08196

Dittman, C. K., Henriquez, A. & Roxburgh, N. (2016). When a non-resident worker is a non-resident parent: Investigating the family impact of fly-in, fly-out work practices in Australia. *Journal of Child and Family Studies*, *25*, 2778–2796. https://doi.org/10.1007/s10826-016-0437-2

*Dorow, S., & Jean, S. (2021). Managing liminal time in the fly-in fly-out work camp. *Human Relations*, *0*(0), 0018726721989792. https://doi.org/10.1177/0018726721989792

Education and Health Standing Committee. (2015). *The impact of FIFO work practices on mental health*. http://www.parliament.wa.gov.au/parliament/commit.nsf/(Report ± Lookup ± by ± Com ± ID)/

2E970A7A4934026448257E67002BF9D1/$file/20150617 ± - ± Final ± Report ± w ± signature ± for ± website.pdf

Egger, M., Smith, G. D. & Phillips, A. N. (1997). Meta-analysis: Principles and procedures. *Bmj*, *315*(7121), 1533–1537. https://doi.org/10.1136/bmj.315.7121.1533

Etzion, D., Eden, D. & Lapidot, Y. (1998). Relief from job stressors and burnout: Reserve service as a respite. *Journal of Applied Psychology*, *83*(4), 577–585.

Eurostats. (2021, April). *Tourism statistics - characteristics of tourism trips*. Retrieved from https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Tourism_statistics_-_characteristics_of_tourism_trips#One_in_nine_trips_was_for_professional_purposes

Francis, J. J., Johnston, M., Robertson, C., Glidewell, L., Entwistle, V., Eccles, M. P. & Grimshaw, J. M. (2010). What is an adequate sample size? Operationalising data saturation for theory-based interview studies. *Psychology and Health*, *25*(10), 1229–1245. https://doi.org/10.1080/08870440903194015

Franklin, A., Barbosa Neves, B., Hookway, N., Patulny, R., Tranter, B. & Jaworski, K. (2018). Towards an understanding of loneliness among Australian men: Gender cultures, embodied expression and the social bases of belonging. *Journal of Sociology*, *55*(1), 1–20. https://doi.org/10.1177/1440783318777309

*Gardner, B., Alfrey, K. L., Vandelanotte, C., & Rebar, A. L. (2018). Mental health and well-being concerns of fly-in fly-out workers and their partners in Australia: A qualitative study. *BMJ Open 2018*;8:e019516. http://dx.doi.org/10.1136/bmjopen-2017-019516

*Gent, V. M. (2004). The impact of fly-in/fly-out work on well-being and work-life satisfaction. Honours thesis, Murdoch University, Western Australia, Australia. Retrieved from https://www.ifap.asn.au/Documents/News%20and%20Media/FIFO_Report_2004.pdf

Geppert, M. & Pastuh, D. (2017). Total institutions revisited: What can goffman's approach tell us about 'oppressive'control and 'problematic'conditions of work and employment in contemporary business organizations? *Competition & Change*, *21*(4), 253–273. https://doi.org/10.1177/1024529417722350

Gilbert, J., Fruhen, L.S., & Parker, S. K. (in prep). Applying Total Institution theory to understanding the Australian FIFO work context.

*Gilbert, J. M. (2019). Between a Rock and a Hard Place: Exploring Ways to Optimise Wellbeing for People Who Work Fly-in, Fly-out in the Australian Mining Industry. Doctoral dissertation, Curtin University.

Government of Western Australia Department of State Development. (2016). Western Australia Economic Profile. Retrieved from http://www.dsd.wa.gov.au/docs/default-source/default-document-library/wa-economic-profile—january-2016?sfvrsn=2

Greenhaus, J. H. & Beutell, N. J. (1985). Sources of conflict between work and family roles. *Academy of Management Review*, *10*(1), 76–88. https://doi.org/10.5465/amr.1985.4277352

Greenspoon, P. J. & Saklofske, D. H. (2001). Toward an integration of subjective well-being and psychopathology. *Social Indicators Research*, *54*, 81–108. https://doi.org/10.1023/A:1007219227883

Gustafson, P. (2012). Managing business travel: Developments and dilemmas in corporate travel management. *Tourism Management*, *33*(2), 276–284. https://doi.org/10.1016/j.tourman.2011.03.006

Hällgren, M., Rouleau, L. & De Rond, M. (2018). A matter of life or death: How extreme context research matters for management and organization studies. *Academy of Management Annals*, *12*(1), 111–153. https://doi.org/10.5465/annals.2016.0017

*Haslam McKenzie, F. M., & Hoath, A. (2016). Aboriginal mine workers: Opportunities and challenges of long-distance commuting. In Haslam McKenzie, F. M (Ed.) *Labour force mobility in the Australian resources industry* (pp. 157–170). Springer.

*Henry, P., Hamilton, K., Watson, S., & Macdonald, N. (2013). FIFO/DIDO mental health research report. Perth, Western Australia: Lifeline WA. Retrieved from http://www.workplacehealth.org.au/_literature_175869/FIFO_DIDO_Mental_Health_Research_Report

Herath, T. & Herath, H. S. (2020). Coping with the new normal imposed by the COVID-19 pandemic: Lessons for technology management and governance. *Information Systems Management*, *37*(4), 277–283. https://doi.org/10.1080/10580530.2020.1818902

Hobfoll, S. E. (1989). Conservation of resources: A new attempt at conceptualizing stress. *American Psychologist*, *44*(3), 513–524. https://psycnet.apa.org/doi/10.1037/0003-066X.44.3.513

Hobfoll, S. E. (2011). Conservation of resource caravans and engaged settings. *Journal of Occupational and Organizational Psychology*, *84*, 116–122. https://doi.org/10.1111/j.2044-8325.2010.02016.x

Hochschild, A. R. (1997). *The time bind: When work becomes home and home becomes work*. Metropolitan.

Hughes, K. D. & Silver, W. A. (2020). Beyond time-binds: Rethinking work–family dynamics for a mobile world. *Human Relations*, *73*(7), 924–952. https://doi.org/10.1177/0018726719846264

Impellizzeri, F. M. & Bizzini, M. (2012). Systematic review and meta-analysis: A primer. *International Journal of Sports Physical Therapy*, *7*(5), 493–503.

Jensen, M. T. (2014). Exploring business travel with work–family conflict and the emotional exhaustion component of burnout as outcome variables: The job demands–resources perspective. *European Journal of Work and Organizational Psychology*, *23*(4), 497–510. https://doi.org/10.1080/1359432X.2013.787183

Johns, G. (2006). The essential impact of context on organizational behavior. *The Academy of Management Review*, *31*(2), 386–408. https://doi.org/10.5465/amr.2006.20208687

Johnsen, C. G. & Sørensen, B. M. (2015). 'It's capitalism on coke!': From temporary to permanent liminality in organization studies. *Culture and Organization*, *21*(4), 321–337. https://doi.org/10.1080/14759551.2014.901326

*Joyce, S. J., Tomlin, S. M., Somerford, P. J., & Weeramanthri, T. S. (2013). Health behaviours and outcomes associated with fly-in fly-out and shift workers in western Australia. *Internal Medicine Journal*, *43*, 440–444. https://doi.org/10.1111/j.1445-5994.2012.02885.x

Kaczmarek, E. A. & Sibbel, A.M. (2008). The psychosocial well-being of children from Australian military and fly-in/fly-out (FIFO) mining families. *Community, Work and Family*, *11*(3), 297–312. https://doi.org/10.1080/13668800801890129

Karasek, R. AJr. (1979). Job demands, job decision latitude, and mental strain: Implications for job redesign. *Administrative Science Quarterly*, *24*(2), 285–308. https://doi.org/10.2307/2392498

Kessler, R. C., Andrews, G., Colpe, L. J., Hiripi, E., Mroczek, D. K. & Normand, S. L., … & Zaslavsky, A. M. (2002). Short screening scales to monitor population prevalences and trends in non-specific psychological distress. *Psychological Medicine*, *32*(6), 959–976. https://doi.org/10.1017/S0033291702006074

*Korneeva, Y., & Simonova, N. (2020). Job stress and working capacity among fly-in-fly-out workers in the oil and gas extraction industries in the Arctic. *International Journal of Environmental Research and Public Health*, *17*(21), 7759. https://doi.org/10.3390/ijerph17217759

Kossek, E. E. & Lautsch, B. A. (2012). Work-family boundary management styles in organizations: A cross-level model. *Organizational Psychology Review*, *2*(2), 152–171. https://doi.org/10.1177/2041386611436264

Kossek, E. E., Lautsch, B. A. & Eaton, S. C. (2005). Flexibility enactment theory: Implications of flexibility type, control, and boundary management for work–family effectiveness. In Kossek, E. E & Lambert's, S. J (Eds.), *Work and life integration: Organizational, cultural, and individual perspectives* (pp. 243–261). Lawrence Erlbaum.

Kreiner, G. E. (2006). Consequences of work-home segmentation or integration: A person-environment fit perspective. *Journal of Organizational Behavior*, *27*, 485–507. https://doi.org/10.1002/job.386

Krippendorff, K. (2004). *Content analysis: An introduction to its methodology*. Thousand Oaks, CA: Sage.

Kuhler, M., & Jelinek, N. (2013). *Autonomy and the Self*. SpringerLink.

Lake, R. (2020). *Business travel statistics: 23 speedy facts to know*. CreditDonkey. Retrieved from: https://www.creditdonkey.com/business-travel-statistics.html

Lamers, S. M. A., Westerhof, G. J., Bohlmeijer, E. T., ten Klooster, P. M. & Keyes, C. L. M. (2011). Evaluating the psychometric properties of the mental health Continuum-short form

(MHC-SF) in the Dutch population. *Journal of Clinical Psychology*, *67*, 99–110. https://doi.org/10.1002/jclp.20741

Lautsch, B. A. & Kossek, E. E. (2012). Work–family boundary management styles in organizations: A cross-level model. *Organizational Psychology Review*, *2*(2), 152–171. https://doi.org/10.1177/2041386611436264

*Lester, L., Waters, S., Spears, B., Epstein, M., Watson, J., & Wenden, E. (2015). Parenting adolescents: Developing strategies for FIFO parents. *Journal of Child and Family Studies*, *24*, 3757–3766. https://doi.org/10.1007/s10826-015-0183-x

Leymann, H. (1996). The content and development of mobbing at work. *European Journal of Work and Organizational Psychology*, *5*(2), 165–184. https://doi.org/10.1080/13594329608414853

Liberati, A., Altman, D. G., Tetzlaff, J., Mulrow, C., Gøtzsche, P. C. & Ioannidis, J. P., … & Moher, D. (2009). The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: Explanation and elaboration. *PLoS Medicine*, *6*, e1000100.

Liese, B., Mundt, K., Dell, L., Nagy, L. & & Demure, B. (1997). Medical insurance claims associated with international business travel. *Occupational and Environmental Medicine*, *54*, 499–503. http://dx.doi.org/10.1136/oem.54.7.499

Lovibond, P. F. & Lovibond, S. H. (1995). The structure of negative emotional states: Comparison of the depression anxiety stress scales (DASS) with the beck depression and anxiety inventories. *Behaviour Research and Therapy*, *33*(3), 335–343. https://doi.org/10.1016/0005-7967(94)00075-U

Lück, D. & Schneider, N. F. (2010). Introduction to the special issue on'Mobility and family': Increasing job mobility-changing family lives. *Zeitschrift für Familienforschung*, *22*(2), 135–148. https://nbn-resolving.org/urn:nbn:de:0168-ssoar-355447

Luthans, F. (2002). The need for and meaning of positive organizational behavior. *Journal of Organizational Behavior: The International Journal of Industrial, Occupational and Organizational Psychology and Behavior*, *23*(6), 695–706. https://doi.org/10.1002/job.165

Luthans, F., Avolio, B. J., Avey, J. B. & Norman, S. M. (2007). Positive psychological capital: Measurement and relationship with performance and satisfaction. *Personnel Psychology*, *60*, 541–572. https://doi.org/10.1111/j.1744-6570.2007.00083.x

Lyons, G. & Chatterjee, K. (2008). A human perspective on the daily commute: Costs, benefits and trade-offs. *Transport Reviews*, *28*(2), 181–198. https://doi.org/10.1080/01441640701559484

Mäkelä, L., Bergbom, B., Tanskanen, J. & Kinnunen, U. (2014). The relationship between international business travel and sleep problems via work-family conflict. *Career Development International*, *19*(7), 794–812. https://doi.org/10.1108/CDI-04-2014-0048

*Martin, J. (2020). FIFO Rosters and workers' health and safety: A case study of the impacts of extended shift rosters on electrical workers in construction in the resources sector. *Labour & Industry: A Journal of the Social and Economic Relations of Work*, *30*(4), 378–400. https://doi.org/10.1080/10301763.2020.1865087

Maslach, C. (1982). *Burnout: The cost of caring*. Prentice-Hall.

*McTernan, W. P., Dollard, M. F., Tuckey, M. R., & Vandenberg, R. J. (2016). Beneath the surface: An exploration of remoteness and work stress in the mines. In Shimazu, A., Bin Nordin, R., Dollard, M. & Oakman, J. (Eds.), *Psychosocial factors at work in the Asia pacific: From theory to practice*, (pp. 341—358). Springer International Publishing.

Meil, G. (2014). Professional mobility in the careers of European workers and division of housework. In Rupp, M., Kapella, O., & Schneider, N. F. (Eds.), *Die Zukunft der Familie. Anforderungen an Familienpolitik und Familienwissenschaft* (pp. 39–52). Verlag Barbara Budrich.

Meyer, J. P., Allen, N. J. & Gellatly, I. R. (1990). Affective and continuance commitment to the organization: Evaluation of measures and analysis of concurrent and time-lagged relations. *Journal of Applied Psychology*, *75*(6), 710–720. https://psycnet.apa.org/doi/10.1037/0021-9010.75.6.710

*Miller, P., Brook, L., Stomski, N., Ditchburn, G., & Morrison, P. (2020). Bullying in fly-in-fly-out employees in the Australian resources sector: A cross-sectional study. *Plos One*, *15*, e0229970.

*Miller, P., Brook, L., Stomski, N. J., Ditchburn, G., & Morrison, P. (2019). Depression, suicide risk, and workplace bullying: A comparative study of fly-in, fly-out and residential resource workers in Australia. *Australian Health Review*, *44*(2), 248–253. https://doi.org/10.1071/AH18155

Misan, G. M. & Rudnik, E. (2015). The pros and cons of long distance commuting: Comments from south Australian mining and resource workers. *Journal of Economic & Social Policy*, *17*(1), 119–157. https://search.informit.org/doi/10.3316/ielapa.243648862587954

Montazer, S., Brumley, K. M. & Maguire, K. (2020). Overnight work-travel, work-to-family conflict, and psychological distress. *The Social Science Journal*, 1–16. https://doi.org/10.1080/03623319.2020.1756175

Murray, C. J. & Lopez, A. D., & World Health Organization. (1996). *The global burden of disease: A comprehensive assessment of mortality and disability from diseases, injuries, and risk factors in 1990 and projected to 2020: summary*. World Health Organization. https://apps.who.int/iris/handle/10665/41864.

Newnham, E. A., Hooke, G. R. & Page, A. C. (2010). Progress monitoring and feedback in psychiatric care reduces depressive symptoms. *Journal of Affective Disorders*, *127*(1-3), 139–146. https://doi.org/10.1016/j.jad.2010.05.003

Nippert-Eng, C. E. (1996). *Home and work: Negotiating boundaries through everyday life*. University of Chicago Press.

Palmer, R. (2014). The money trail: An exploration of perspectives on money and materialism in FIFO employment. In Brueckner, M., Durey, A., Mayes, R. & Pforr, C. (Eds). *Resource curse or cure?* (pp. 107–119). Springer.

*Parker, S., Fruhen, L., Burton, C., McQuade, S., Loveny, J., Griffin, M., Page, A., Chikritzhs, T., Crock, S., Jorritsma, K., & Esmond, J. (2018). Impact of FIFO work arrangements on the mental health and wellbeing of FIFO workers. Centre for Transformative Work Design. https://www.mhc.wa.gov.au/media/2547/impact-of-fifo-work-arrangement-on-the-mental-health-and-wellbeing-of-fifo-workers-full-report.pdf

Parkes, K. R. (2015). Shift rotation, overtime, age, and anxiety as predictors of offshore sleep patterns. *Journal of Occupational Health Psychology*, *20*(1), 27–39. https://psycnet.apa.org/doi/10.1037/a0038164

Parkes, K. R., Carnell, S. C. & Farmer, E. L. (2005). 'Living two lives' perceptions, attitudes and experiences of spouses of UK offshore workers. *Community, Work and Family*, *8*(4), 413–437. https://doi.org/10.1080/13668800500251755

Peck, J. (2013). Polanyi in the pilbara. *Australian Geographer*, *44*(3), 243–264. https://doi.org/10.1080/00049182.2013.817037

Perlman, D. & Peplau, L. A. (1981) Toward a Social Psychology of Loneliness. In Duck, R. & Gilmour, R. (Eds.), *Personal Relationships in Disorder* (pp. 31–56). Academic Press.

*Perring, A., Pham, K., Snow, S., & Buys, L. (2014). Investigation into the effect of infrastructure on fly-in fly-out mining workers. *Australian Journal of Rural Health*, *22*, 323–327. https://doi.org/10.1111/ajr.12117

Petrou, P. & Bakker, A. B. (2016). Crafting one's leisure time in response to high job strain. *Human Relations*, *69*(7), 507–529. https://doi.org/10.1177/0018726715590453

*Pirotta, J. (2009). An exploration of the experiences of women who FIFO. *The Australian Community Psychologist*, *21*(2), 37–51.

Ravalet, E., Vincent-Geslin, S. & Dubois, Y. (2017). Job-related "high mobility" in times of economic crisis: Analysis from four European countries. *Journal of Urban Affairs*, *39*(4), 563–580. https://doi.org/10.1080/07352166.2016.1251170

Restifo, E., Kashyap, S., Hooke, G. R. & Page, A. C. (2015). Daily monitoring of temporal trajectories of suicidal ideation predict self-injury: A novel application. *Psychotherapy Research*, *25*(6), 705–713. https://doi.org/10.1080/10503307.2015.1006707

Rogers, H. L. & Reilly, S. M. (2002). A survey of the health experiences of international business travelers. *Journal of the American Association of Occupational Health Nurses*, *50*(10), 449–459. https://doi.org/10.1177/216507990205001006

Rüger, H., Feldhaus, M., Becker, K. S. & Schlegel, M. (2011). Circular job-related spatial mobility in Germany: Comparative analyses of two representative surveys on the forms, prevalence and relevance in the context of partnership and family development.

Comparative Population Studies, *36*(1), 221–248. https://doi.org/10.12765/CPoS-2011-05.

Rüger, H., Pfaff, S., Skora, T. & Schneider, N. F. (2016). *Job Mobilities and Family Lives in Europe – Second Wave: Panel Data Set & Oversampling. BiB Daten- und Methodenberichte 3/2016*. Bundesinstitut für Bevölkerungsforschung.

Rüger, H. & Viry, G. (2017). Work-related travel over the life course and its link to fertility: A comparison between four European countries. *European Sociological Review*, *33*(5), 645–660. https://doi.org/10.1093/esr/jcx064

Rundle, A. G., Revenson, T. A. & Friedman, M. (2018). Business travel and behavioral and mental health. *Journal of Occupational and Environmental Medicine*, *60*(7), 612–616. https://doi.org/10.1097/JOM.0000000000001262

Sarason, S. B. (1974). *The psychological sense of community: Prospects for a community psychology*. California: Jossey-Bass Inc. Publishers.

*Saxinger, G. (2021). Rootedness along the way: Meaningful sociality in petroleum and mining mobile worker camps. *Mobilities*, *16*(2), 194–211. https://doi.org/10.1080/17450101.2021.1885844

*Sellenger, M., Oosthuizen, J. (2017). Quantitative analysis of mental wellbeing of fly-in fly-out construction project support service workers. *Journal of Preventive Medicine and Healthcare*, *1*(1), 1–6.

Shockley, K. M. & Allen, T. D. (2007). When flexibility helps: Another look at the availability of flexible work arrangements and work–family conflict. *Journal of Vocational Behavior*, *71*(3), 479–493. https://doi.org/10.1016/j.jvb.2007.08.006

*Sibbel, A. M. (2010). Living FIFO: The experiences and psychosocial wellbeing of Western Australian fly-in/fly-out employees and partners. PhD thesis, Edith Cowan University, Western Australia, Australia. Retrieved from http://ro.ecu.edu.au/theses/132/

*Sibbel, A. M., Kaczmarek, E., & Drake, D. (2016). Fly-in/fly-out accommodation: Workers' perspectives. In *Labour force mobility in the Australian resources industry* (pp. 137–156). Springer.

Sonnentag, S. & Bayer, U.-V. (2005). Switching off mentally: Predictors and consequences of psychological detachment from work during off-job time. *Journal of Occupational Health Psychology*, *10*(4), 393–414. https://psycnet.apa.org/doi/10.1037/1076-8998.10.4.393

Sonnentag, S. & Fritz, C. (2007). The recovery experience questionnaire: Development and validation of a measure for assessing recuperation and unwinding from work. *Journal of Occupational Health Psychology*, *12*(3), 204–221. https://psycnet.apa.org/doi/10.1037/1076-8998.12.3.204

Stanko, T. L. & Beckman, C. M. (2015). Watching you watching me: Boundary control and capturing attention in the context of ubiquitous technology use. *Academy of Management Journal*, *58*(3), 712–738. https://doi.org/10.5465/amj.2012.0911

Storey, K. (2001). Fly-in/fly-out and fly-over: Mining and regional development in western Australia. *Australian Geographer*, *32*(2), 133–148. https://doi.org/10.1080/00049180120066616

Striker, J., Dimberg, L. & Liese, B. H. (2000). Stress and business travel: Individual, managerial, and corporate concerns. *Journal of Organizational Excellence*, *20*, 3–10. https://doi.org/10.1002/1520-6734(200024)20:1%3C3::AID-NPR2%3E3.0.CO;2-U

*Sutherland, R. C., Chur-Hansen, A., & Winefield, H. (2017). Experiences of fly-in, fly-out and drive-in, drive-out rural and remote psychologists. *Australian Psychologist*, *52*, 219–229. https://doi.org/10.1111/ap.12194

Taylor, J. C. & Simmonds, J. G. (2009). Family stress and coping in the fly-in fly-out workforce. *The Australian Community Psychologist*, *21*(2), 23–36. Retrieved from https://groups.psychology.org.au/Assets/Files/Taylor-21(2)-2009.pdf

*Torkington, A. M., Larkins, S., & Gupta, T. S. (2011). The psychosocial impacts of fly-in fly-out and drive-in drive-out mining on mining employees: A qualitative study. *Australian Journal of Rural Health*, *19*, 135–141. https://doi.org/10.1111/j.1440-1584.2011.01205.x

*Tuck, J., Temple, E. C., & Sipek, M. (2013). Wellbeing of fly-in/fly-out and drive-in/drive-out employees: Evidence from Australia. 6th International Conference on Sustainable Development in the Minerals Industry, Milos Island, Greece, 30 June–3 July 2013. Ballarat: University of Ballarat.

VanderWeele, T.Y., Hawkley, L.C. & Cacioppo, J.T. (2012) On the reciprocal association between loneliness and subjective well-being. *American Journal of Epidemiology*, *176*(1), 777–784. https://doi.org/10.1093/aje/kws173

*Velander, F., Schineanu, A., WenBin, L., & Richard, M. (2010). Digging for gold and coming up blue: A health survey in the mining industry. *Australian and New Zealand Journal of Health, Safety and Environment*, *26*(5), 389–401.

Viry, G., Ravalet, E. & Kaufmann, V. (2015). High mobility in Europe: An overview. In Viry, G. & Kaufmann, V. (Eds). *High mobility in Europe: Work and personal life*. pp. 29–58. Palgrave Macmillan.

*Vojnovic, P., & Bahn, S. (2015). Depression, anxiety and stress symptoms among FIFO workers. *Journal of Health Safety and Environment*, *31*(3), 207–223.

Vojnovic, P., Michelson, G., Jackson, D. & Bahn, S. (2014). Adjustment, well-being and help-seeking among Australian FIFO mining employees. *Australian Bulletin of Labour*, *40*(2), 242–261. https://search.informit.org/doi/10.3316/informit.791370738447216

Voydanoff, P. (2005). Toward a conceptualization of perceived work-family fit and balance: A demands and resources approach. *Journal of Marriage and Family*, *67*, 822–836. https://doi.org/10.1111/j.1741-3737.2005.00178.x

*Watts, J. (2004). *Best of both worlds? Fly In–Fly Out research project final report*. Pilbara Regional Council. Retrieved from http://inform.regionalaustralia.org.au/process/community-engagement/item/best-of-both-worlds

Wood, G., Hilti, N., Kramer, C. & Schier, M. (2015). A residential perspective on multi-locality. *Tijdschrift Voor Economische En Sociale Geografie*, *106*(4), 363–377. https://doi.org/10.1111/tesg.12158

World Health Organisation. (2013). Comprehensive Mental Health Action Plan. Retrieved from http://www.who.int/mental_health/action_plan_2013/en/

Author biographies

Laura S. Fruhen is a Senior Lecturer of Applied Psychology at the University of Western Australia. She investigates leadership, teamwork, work design, culture, and attitudes and links these with health and safety, well-being, and other outcomes.

Jess Gilbert is a Postdoctoral Research Fellow at Curtin University's Future of Work Institute. She researches mental health at work, particularly in atypical work designs, community psychology, and LGBTQIA+ people and experiences of mental health.

Sharon K. Parker is an Australian Research Council Laureate Fellow, a distinguished Professor of Organisational Behavior at Curtin University and an Honorary Professor at the University of Sheffield. Her research focuses particularly on job and work design, and she is also interested in proactive behavior, change, wellbeing, development, and job performance.